COMSM0067: Advanced Topics in Programming Languages
Problem Sheet 8

Hoare Logic I — triples and the rules

Throughout, $a > a'$ in a program abbreviates the IMP boolean $\neg(a \leq a')$.

1. Draw derivations that justify the following triples, naming every rule used.

   (i) $\{x = 3\}\ \ y := x + 1\ \ \{y = 4\}$

   (ii) $\{\mathsf{True}\}\ \ x := 1;\ y := x + 1\ \ \{y = 2\}$

   (iii) $\{x = A\}\ \ \mathsf{if}\ x \leq 0\ \mathsf{then}\ y := -x\ \mathsf{else}\ y := x\ \ \{y = |A|\}$

   (iv) $\{\mathsf{False}\}\ \ x := x + 1\ \ \{x = 100\}$

2. (*) Prove that the following program meets its specification. There is no loop, so no invariant is needed: start from the postcondition and push it backwards through each branch with ASSIGN and SEQ, writing down the assertion you obtain at each step, then use IF and CONSEQ to finish.

   $$\{x = A \wedge y = B\}\ \ \mathsf{if}\ x > y\ \mathsf{then}\ (\, t := x;\ x := y;\ y := t\,)\ \mathsf{else\ skip}\ \ \{x \leq y \wedge x + y = A + B\}$$

   Say which rule you use to bring the two branches of the if to a common postcondition.